

Project Workflow: Week 4

Project: EasyStego: A Drag-and-Drop Image Steganography Tool

Phase: GUI Layout: Tabs, Drop Zones & Theming

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Course: Software Development I (CSE 2216)

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1. Objective

The primary goal of Week 4 was to build the full graphical user interface layout of the EasyStego application. This included:

- Building the JTabbedPane with two distinct tabs: Hide Message and Reveal Message.
- Implementing the dashed-border drop zone label for image input in both tabs.
- Adding the message text area, capacity status label, and action buttons to each tab.
- Applying the application-wide visual theme: Segoe UI font, Windows-blue accent color (#0078D7), and drop-zone background color.
- Verifying the complete GUI layout renders correctly before logic is added in Week 5 and 6.

2. Detailed Task Breakdown

JTabbedPane Setup: Created a JTabbedPane as the central container inside the JFrame. Added two tabs using addTab(): the first labeled '1. Hide Message' and the second '2. Reveal Message'. Applied the TITLE_FONT (Segoe UI Bold 16px) to the tab bar for consistent styling.

Hide Message Tab Layout: Constructed the Hide Message panel using BorderLayout with 15px horizontal and vertical gaps. The top section holds the dashed drop zone (200px height). The center section contains a message label, JTextArea wrapped in JScrollPane, and a capacity label. The bottom section holds the Encrypt & Save Image action button styled with the Windows-blue accent color (#0078D7).

Reveal Message Tab Layout: Built the Reveal Message panel with the same BorderLayout structure. The top section holds the stego-image drop zone. The center section contains the Hidden Message label and a read-only monospaced JTextArea. The bottom section holds the Copy Text to Clipboard button styled in green (#2E8B57).

Drop Zone Implementation: Created a reusable createDropZone() helper method that returns a styled JLabel. The label uses SwingConstants.CENTER alignment, Segoe UI Bold font, a light gray background (#F0F0F0), and a dashed border created with BorderFactory.createDashedBorder(). A CompoundBorder adds 20px inner padding around the dashed border.

Global Theming: Defined three final constants at the class level: MAIN_FONT (Segoe UI Plain 14), TITLE_FONT (Segoe UI Bold 16), and ACCENT_COLOR (RGB 0, 120, 215 — Windows blue). Applied UIManager.setLookAndFeel(getSystemLookAndFeelClassName()) to match the

native OS appearance. Created a reusable `createStyledButton()` helper that sets background color, white foreground, removes focus paint, and fixes button height at 50px.

Layout Verification: Compiled and ran the updated `EasyStego.java` to visually verify that both tabs render correctly, the drop zones display their placeholder text, the text areas scroll properly, and the styled buttons appear at the correct size and color before any logic is wired in Week 5.

3. GUI Components Added This Week

Component	Class / Method	Location
Tabbed container	<code>JTabbedPane</code>	<code>EasyStego</code> constructor
Hide Message panel	<code>createHidePanel()</code>	Tab 1
Reveal Message panel	<code>createRevealPanel()</code>	Tab 2
Drop zone label	<code>createDropZone()</code>	Both tabs
Message input area	<code>JTextArea</code> + <code>JScrollPane</code>	Hide tab center
Capacity status label	<code>JLabel</code> (<code>capacityLabel</code>)	Hide tab center
Output message area	<code>JTextArea</code> (read-only)	Reveal tab center
Encrypt & Save button	<code>createStyledButton()</code>	Hide tab bottom
Copy to Clipboard button	<code>createStyledButton()</code>	Reveal tab bottom

4. Deliverables

- Updated `EasyStego.java`: full GUI layout with `JTabbedPane`, both tab panels, drop zones, text areas, capacity label, and styled buttons.
- `createHidePanel()` method: complete Hide Message tab layout.
- `createRevealPanel()` method: complete Reveal Message tab layout.
- `createDropZone()` helper: reusable dashed-border drop zone label.
- `createStyledButton()` helper: reusable themed action button.
- Verified application window rendering both tabs correctly with no logic errors.
- Updated commit pushed to GitHub.

5. Tools & Resources

Development Environment:

- Java JDK 11+ — compiler and runtime
- Visual Studio Code with Java Extension Pack
- Git / GitHub — version control

Java Swing Classes Used:

- `javax.swing.JTabbedPane` — tab container

- javax.swing.JPanel with BorderLayout — tab panel structure
- javax.swing.JLabel — drop zone and capacity display
- javax.swing.JTextArea + JScrollPane — message input and output
- javax.swing.BorderFactory — dashed border and compound border
- javax.swing.JButton — styled action buttons
- javax.swing.UIManager — system look and feel